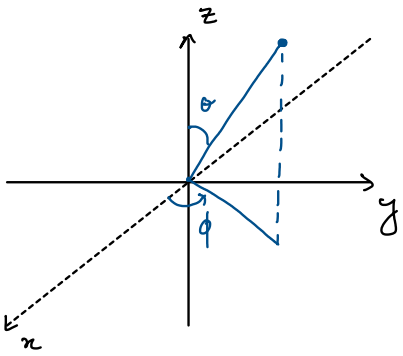
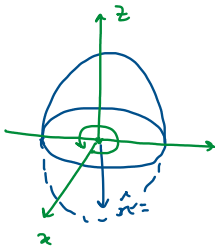
Week 3

Monday, 4 September 2023 9:54 PM

Spherical Coordinates





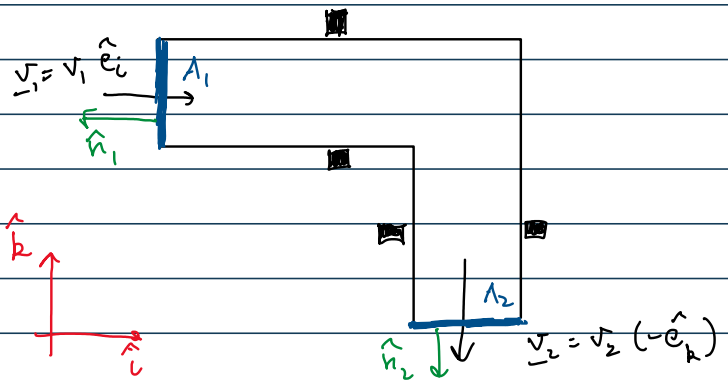
$0 < \phi < 2\pi$
$0 < \theta < \pi$
} Full sphere

$-\frac{\pi}{2}$

L6 Surface Tension, Reynolds Transport Theorem

Monday, 4 September 2023 10:11 PM

L6 Surface Tension, ...



+ Constant velocity
+ Normal vectors on each surface/opening

$\hat{n}_1 = -\hat{e}_i$

$\hat{n}_2 = -\hat{e}_k$

→ Momentum RTT:

w/ no s.s. $\quad \Sigma F = \iint_A \rho \underline{v} (\underline{v} \cdot \hat{n}) \, dA$

$$\Sigma F = \rho \underline{v}_1 \iint_{A_1} [(v_1 \hat{e}_i) \cdot (-\hat{e}_i)] \, dA + \rho \underline{v}_2 \iint_{A_2} [v_2 (-\hat{e}_k) \cdot (-\hat{e}_k)] \, dA$$

$\underline{v}_1 = v_1 \hat{e}_i$, $\hat{n}_1 = -\hat{e}_i$, $\hat{e}_i \cdot \hat{e}_i = 1$; $\underline{v}_2 = v_2 (-\hat{e}_k)$, $\hat{n}_2 = -\hat{e}_k$, $\hat{e}_k \cdot \hat{e}_k = 1$

$$= \rho (v_1 \hat{e}_i)(-v_1) \iint_{A_1} dA + \rho (v_2 (-\hat{e}_k)) v_2 \iint_{A_2} dA$$

$$\boxed{\Sigma F \Rightarrow -\rho v_1^2 A_1 \hat{e}_i - \rho v_2^2 A_2 \hat{e}_k}$$

7.

Derivation

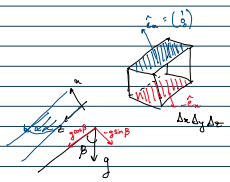
Thursday, 21 September 2023 6:32 PM

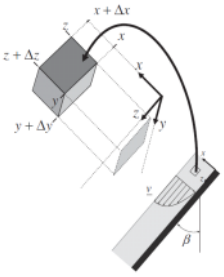
Inclined plane Derivation

$$f = \int\int_A (\hat{n}\cdot\underline{\underline{\Pi}})\,dS$$

$$(\hat{n}\cdot\underline{\underline{\Pi}})\Big|_{top} = \underset{1\times3}{(1\;0\;0)}\begin{pmatrix} \tau_{xx}-p & \tau_{xy} & \tau_{xz} \\ \vdots & \ddots & \vdots \\ \cdots & \cdots & \cdots \end{pmatrix}_{3\times3}$$

$$= (\tau_{xx}-p \quad \tau_{xy} \quad \tau_{xz})$$





$$\begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}_{3\times1}\underset{1\times3}{(v_x\;\;v_y\;\;v_z)} = \begin{pmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} \\ \frac{\partial v_x}{\partial z} & \frac{\partial v_y}{\partial z} & \frac{\partial v_z}{\partial z} \end{pmatrix}$$

$$\sum_{on\ CV} f = 0 = f_{gravity} + \sum_{i=1}^{6} f_{i^{th}\ side}$$

↳ force balance on a fluid element

$$\sum f = \begin{pmatrix} -\rho g\sin\beta \\ 0 \\ \rho g\cos\beta \end{pmatrix}\Delta x\Delta y\Delta z + \left[\Delta y\Delta z\begin{pmatrix} \tau_{xx}-p \\ \tau_{xy} \\ \tau_{xz} \end{pmatrix}_{xyz}\Big|_{x+\Delta x} - \Delta y\Delta z\begin{pmatrix} \tau_{xx}-p \\ \tau_{xy} \\ \tau_{xz} \end{pmatrix}_{xyz}\Big|_{x} \cdots\cdots\right]$$

↳ Split into each x, y, z component & taking limits

$$x: \quad 0 = -\rho g\sin\beta + \left(\frac{\tau_{xx}|_{x+\Delta x} - \tau_{xx}|_x}{\Delta x}\right) - \left(\frac{p|_{x+\Delta x} - p}{\Delta x}\right) + \left(\frac{\tau_{yx}|_{y+\Delta y} - \tau_{yx}|_y}{\Delta y}\right) + \left(\frac{\tau_{zx}|_{z+\Delta z} - \tau_{zx}|_z}{\Delta z}\right)$$

↳ Limit:

$$x: \quad \boxed{0 = -\frac{\partial p}{\partial x} + \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right] - \rho g\sin\beta}$$

x component momentum balance!

[4.357, 4.358, 4.359]

↓

But our goal is to find velocity profile!

$$\tau = \mu\left(\nabla\underline{v} + (\nabla\underline{v})^T\right)$$

↳ Need to use an equation which now relates τ & v → Newtonian constitutive equation

$$v = \begin{pmatrix} 0 \\ 0 \\ v_z(x) \end{pmatrix} \quad ; \quad \underline{\tau} = \mu\begin{pmatrix} 2\frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x}+\frac{\partial v_x}{\partial y} & \frac{\partial v_z}{\partial x}+\frac{\partial v_x}{\partial z} \\ \frac{\partial v_x}{\partial y}+\frac{\partial v_y}{\partial x} & 2\frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y}+\frac{\partial v_y}{\partial z} \\ \frac{\partial v_x}{\partial z}+\frac{\partial v_z}{\partial x} & \frac{\partial v_y}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{xyz}$$

$$\tau = \mu\begin{pmatrix} 0 & 0 & \frac{\partial v_z}{\partial x} \\ 0 & 0 & 0 \\ \frac{\partial v_x}{\partial x} & 0 & 0 \end{pmatrix}_{xyz}$$

(τ_{xx} at top-left, τ_{zy} at bottom-middle)

$$\tau_{xz} = \mu\frac{dv_z}{dx}$$

$$\tau_{zx} = \mu\frac{dv_z}{dx}$$

↓

Substitute in balance to arrive at simplified form

There will be only 1 direction where velocity would remain for unidirectional flows!

$$z: \quad \boxed{0 = -\frac{\partial p}{\partial z} + \mu\frac{d^2v_z}{dx^2} + \rho g\cos\beta}$$

↓

Solve for v_z w/ BC's:

@ $x = 0$

@ $x = H$; $\tau_{xz} = 0 = \mu\frac{dv_z}{dx}$

$$\boxed{v_z(x) = \frac{\rho g\cos\beta}{\mu}\left[Hx - \frac{x^2}{2}\right]}$$

Pg 380, 381, 382

Derivation

Friday, 29 September 2023 12:37 PM

$$\nabla = \left(\hat{e}_r \frac{\partial}{\partial r} + \frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta} + \hat{e}_z \frac{\partial}{\partial z} \right) \quad ; \quad \underline{v} = \left(v_r \hat{e}_r + v_\theta \hat{e}_\theta + v_z \hat{e}_z \right) \quad ; \quad \underline{\underline{\tau}} = \nabla \underline{v} + (\nabla \underline{v})^T = \begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ & \tau_{\theta\theta} & \tau_{\theta z} \\ & & \tau_{zz} \end{pmatrix}$$

$\nabla \underline{v}$ → This is a dyadic product but w/ del operator

↳ $$(\nabla \underline{v})^T = \begin{pmatrix} \hat{e}_r \frac{\partial}{\partial r}(v_r \hat{e}_r) & \hat{e}_r \frac{\partial}{\partial r}(v_\theta \hat{e}_\theta) & \hat{e}_r \frac{\partial}{\partial r}(v_z \hat{e}_z) \\ \frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta}(v_r \hat{e}_r) & \frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta}(v_\theta \hat{e}_\theta) & \frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta}(v_z \hat{e}_z) \\ \hat{e}_z \frac{\partial}{\partial z}(v_r \hat{e}_r) & \hat{e}_z \frac{\partial}{\partial z}(v_\theta \hat{e}_\theta) & \hat{e}_z \frac{\partial}{\partial z}(v_z \hat{e}_z) \end{pmatrix}$$

(1) $$\hat{e}_r \left[\frac{\partial}{\partial r}(v_r \hat{e}_r) \right] = \hat{e}_r \left[\frac{\partial v_r}{\partial r} \hat{e}_r \right] + v_r \cancel{\frac{\partial \hat{e}_r}{\partial r}}^{0} = \frac{\partial v_r}{\partial r} \hat{e}_r \hat{e}_r \quad ; \quad \frac{\partial \hat{e}_r}{\partial r} = \frac{\partial \hat{e}_\theta}{\partial r} = \frac{\partial \hat{e}_z}{\partial r} = 0 \quad \& \quad \frac{\partial \hat{e}_r}{\partial \theta} = \hat{e}_\theta \; ; \; \frac{\partial \hat{e}_\theta}{\partial \theta} = -\hat{e}_r$$

Similarly,

(2) $$\frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta}(v_\theta \hat{e}_\theta) = \frac{\hat{e}_\theta}{r} \left[\left(\hat{e}_\theta \frac{\partial v_\theta}{\partial \theta} \right) + v_\theta \cancel{\frac{\partial \hat{e}_\theta}{\partial \theta}}^{-\hat{e}_r} \right] = \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} \hat{e}_\theta \hat{e}_\theta - \frac{1}{r} v_\theta \hat{e}_\theta \hat{e}_r$$

$$\hat{e}_r \frac{\partial}{\partial r}(v_\theta \hat{e}_\theta) = \hat{e}_r \left[\hat{e}_\theta \frac{\partial v_\theta}{\partial r} + v_\theta \cancel{\frac{\partial \hat{e}_\theta}{\partial r}}^{0} \right] = \frac{\partial v_\theta}{\partial r} \hat{e}_\theta \hat{e}_r$$

(3) $$\frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta}(v_r \hat{e}_r) = \frac{\hat{e}_\theta}{r} \left[\frac{\partial v_r}{\partial \theta} \hat{e}_r + v_r \cancel{\frac{\partial \hat{e}_r}{\partial \theta}}^{\hat{e}_\theta} \right] = \frac{1}{r} \frac{\partial v_r}{\partial \theta} \hat{e}_r \hat{e}_\theta + \frac{v_r}{r} \hat{e}_\theta \hat{e}_\theta$$

$$(\nabla \underline{v})^T = \begin{pmatrix} \frac{\partial v_r}{\partial r} & \frac{\partial v_\theta}{\partial r} & \frac{\partial v_z}{\partial r} \\ \frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} & \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} \\ \frac{\partial v_r}{\partial z} & \frac{\partial v_\theta}{\partial z} & \frac{\partial v_z}{\partial z} \end{pmatrix}$$

$$\underline{\underline{\tau}} = \nabla \underline{v} + (\nabla \underline{v})^T = \begin{pmatrix} \frac{\partial v_r}{\partial r} & \frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} & \frac{\partial v_r}{\partial z} \\ \frac{\partial v_\theta}{\partial r} & \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} & \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} & \frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z} + \begin{pmatrix} \frac{\partial v_r}{\partial r} & \frac{\partial v_\theta}{\partial r} & \frac{\partial v_z}{\partial r} \\ \frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} & \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} \\ \frac{\partial v_r}{\partial z} & \frac{\partial v_\theta}{\partial z} & \frac{\partial v_z}{\partial z} \end{pmatrix}$$

$$\underline{\underline{\tau}} = \begin{pmatrix} 2\frac{\partial v_r}{\partial r} & \frac{\partial v_\theta}{\partial r} + \frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ & & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

FL Lab2

Friday, 29 September 2023 10:57 PM

Title: Turbulent flow in a pipe

Goal: Determine the friction factor for turbulent flow over a range of flow rates

Theory: Reynold's Number

$$Re = \frac{\rho \langle v \rangle D}{\mu}$$

(ρ: fluid; $\langle v \rangle$: average velocity; D: pipe diameter (ID); μ: viscosity)

$Re > 4000$ (turbulent flow)

(Darcy) Friction Factor (f_D): Twists, turns, pipe flow → frictional losses → can become important in many industrial applications so important to study!

↳ Empirically determined correlations exist

Blasius

$$f_D = 0.316\, Re^{-0.25}$$

Colebrook (surface roughness)

$$\frac{1}{f_D^{1/2}} = -2 \log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re\, f_D^{1/2}}\right)$$

f_D is non-dimensional!

P.S. f → fanning friction factor $f = \frac{1}{4} f_D$ } Same concept but developed by different scientist so off by a factor!

Darcy-Weisbach Equation

Relates head loss due to friction:

$$\frac{\Delta p}{L} = \frac{f_D \cdot \rho \langle v \rangle^2}{2D} = f_D \left(\rho \frac{\langle v \rangle D}{\mu}\right) \frac{\langle v \rangle}{2D^2} \mu = \frac{f_D \langle v \rangle \mu}{2D^2} (Re)$$

Experiment: + Flow water of specific Q → Record variables in order to calculate Re → Calculate f & compare Darcy vs. (Colebrook & Blasius)

Re: D, $\langle v \rangle$, ρ, μ

Fluid properties: water ✓
Pipe specs: given ✓

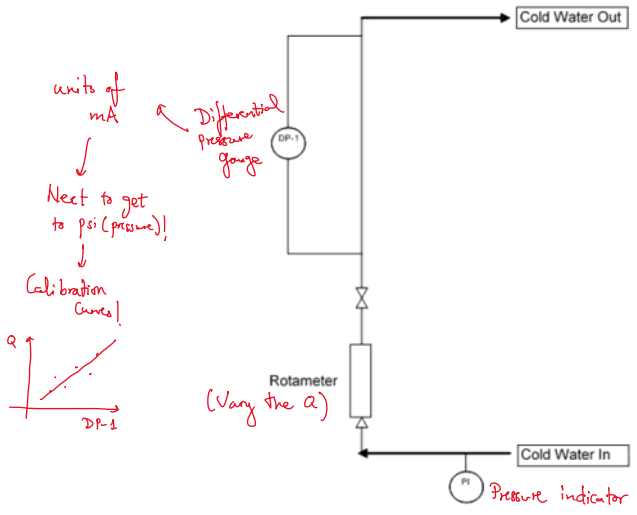
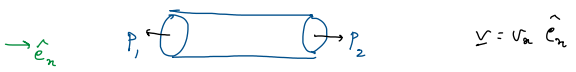


Figure 1: Process Flow Diagram for Turbulent Flow in a Pipe

Pre-Lab (due w/ the lab report due date)

Pre-Lab (due w/ the lab report due date)

Hints: ① → RTT on fluid flow in pipe to evaluate $\sum_{cv} f$
→ $\sum_{cv} f$ balance (pressures + shear stress)



$$\sum_{\text{on fluid}} F = \frac{dP}{dt} + \iint_{CS} \rho(\hat{n} \cdot \underline{v})\, \underline{v}\, dS$$

↓ @ SS & CS as opening & exit

$$\sum_{\text{on fluid}} F = \text{Term 1} + \text{Term 2} = 0$$

If $\Sigma F = 0$; the pressures must be balanced w/ shear stress to give net force = 0

$P_1 A - P_2 A - \tau$ (Surface area)

↓

rearrange to give τ & Δp relation

② Similar analysis



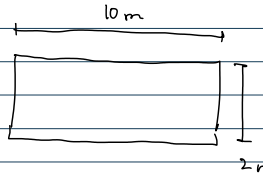
redo ① analysis ; now account for gravity as force in force balance

③ Substitution of τ from ①

④ Substitution & analysis

Tuesday, 28 November 2023 10:16 AM

Sliced flat plate:



$Re_L = \frac{\rho V_\infty L}{\mu} = \frac{(1000)(5)(10)}{(0.001)}$

$= 5 \times 10^7$ (turbulent)

For a BL:

Eqn $C_D = \frac{F_D}{(KE)(area)}$ (No laminar flow contr.)

$\frac{0.031}{(Re_L)^{1/7}} = \frac{F_D}{\frac{1}{2}\rho V_\infty^2 \cdot (20)}$

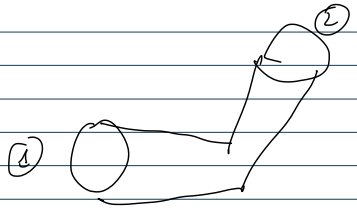
$F_D = \frac{(0.031)((0.5)(1000)\cdot 5^2 \cdot 20)}{(5\times10^7)}$

$F_D = 1.55 \times 10^{-4}$ N

$\times 2$ 3.1×10^{-4}

Atleast ↗

Q2)



Mass balance:

@SS $\rho_1 \langle v_1 \rangle A_1 = \rho_2 \langle v_2 \rangle A_2$; $\rho_1 = \rho_2$

$1(3) = \langle v_2 \rangle 1$

$\langle v_2 \rangle = 3$

MEB: $\frac{\Delta P}{\rho} + \frac{\Delta \langle v \rangle^2}{2} + g\Delta z + F = \frac{W_s}{m}$ (0, 0)

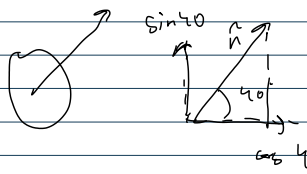
$\frac{P_2 - P_1}{\rho} + \frac{V_2^2 - V_1^2}{2} = 0$ $\alpha = 1$ (turbulent)

$\frac{P_2 - P_1}{2} = -\left(\frac{3^2 - 1^2}{2}\right)$ 9-1

$1 \times 10^5 - P_1 = -4$

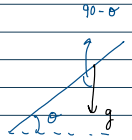
$P_1 = 8 + (1 \times 10^5)$ kg

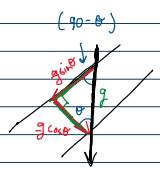
(c)



$\beta = 1$ (turbulent) (Assuming no friction)

(d) Momentum: $\sum \underline{F} = \rho \cos\theta \frac{(V_1)^2}{\beta} A_1 \cdot \hat{e}_1 + \rho \cos\theta \frac{(V_2)^2}{\beta} A_2 \cdot \hat{e}_2$





SOH CAH TOA

$\sin\theta = \frac{opp}{hyp}$

$90 + (90-\theta) + \alpha = 180$

$\alpha = \theta$



$g = -g\sin\theta\,\hat{e}_z + g\cos\theta\,\hat{e}_x$

Q1]



$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}$; $\underline{g} = \begin{pmatrix} 0 \\ 0 \\ g_z \end{pmatrix}$

Effect of gravity ✗

pressure ?



Mass: $\frac{\partial v_z}{\partial z} = 0$

Assumptions

+ Newtonian fluid

Stokes: $\mu \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \rho g_z = 0$

$\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) = -\rho g_z \frac{(r)}{\mu}$

$r\frac{\partial v_z}{\partial r} = \frac{-\rho g_z}{\mu}\frac{r^2}{2} + \frac{C_1}{r}$

$v_z = -\frac{\rho g_z}{2\mu}\left(\frac{r^2}{2}\right) + C_1 \ln r + C_2$

@ $r = R$, $v_z = 0$

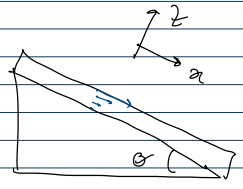
$C_2 = \left(+\frac{\rho g_z R^2}{4\mu} - C_1 \ln R\right)$

@ $r = d+R$, $\tau_{rz} = 0$; $\tau_{rz} = \mu\left(\frac{\partial v_r}{\partial z}^{0} + \frac{\partial v_z}{\partial r}\right) = 0$

$C_1 = \left(\frac{\rho g_z}{2\mu}(d+R)\right)(d+R)$

(i) $v_z\,(@\,r = R)$

Q2]

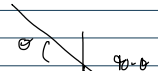


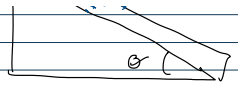
(a) $\nabla \cdot \underline{v} = 0$

$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y}^{0} + \frac{\partial v_z}{\partial z}^{0} = 0$

$v = \begin{pmatrix} v_x \\ 0 \\ v_z \end{pmatrix}$ $p = ?$

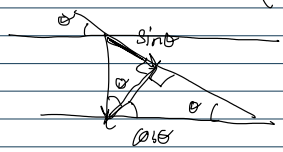
$g = \begin{pmatrix} g\sin\theta \\ 0 \\ -g\cos\theta \end{pmatrix}$

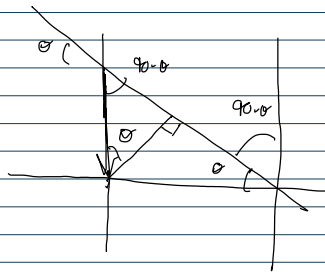




$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

$$g = \begin{pmatrix} g\sin\theta \\ 0 \\ -g\cos\theta \end{pmatrix}$$





(b)

$$0 = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 v_x}{\partial z^2}\right) + \rho g_x$$

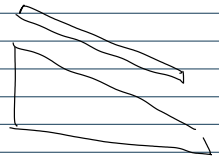
$$0 = -\frac{\partial p}{\partial y} 0$$

$$0 = -\frac{\partial p}{\partial z} + \mu\left(\frac{\partial^2}{\partial z^2}\right) + \rho g_z$$

(c) @ $z = 0$, $v_x = 0$

@ $z = H$, $\tau_{zx} = 0$; $\tau_{zz} =$

(d)



@ $z = H$, $v_x = 0$

Problem Clarification

Friday, 27 October 2023 9:58 AM

$$\underline{F} = \iint_{@surface} (\hat{n} \cdot \underline{\underline{\Pi}}) \, dA$$

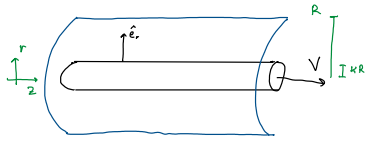
- $\hat{n}$: normal to surface
- $\underline{F}$: Force on a surface of area given by dA
- $\underline{\underline{\Pi}}$: Total stress tensor contains information about the stress at a point in a fluid

[Force on a surface (by the fluid flow) is positive]

∴ 1) Bounds on integral must define the area of the surface (not the fluid!)

2) The integral is evaluated @ the surface coordinates (i.e. the jacobian will be at specific coordinate)

eg:



+ Force on wire:

$$\underline{F} = \iint_{cs} (\hat{n} \cdot \underline{\underline{\Pi}}) \, dS$$

surface is the entire wire / small cylinder

where:

$$0 < \theta < 2\pi$$
$$0 < z < L$$

$$\therefore \underline{F} = \int_0^L \int_0^{2\pi} \left[\hat{e}_r \cdot \begin{pmatrix} -p & 0 & \mu \frac{\partial v_z}{\partial r} \\ 0 & -p & 0 \\ \mu \frac{\partial v_z}{\partial r} & 0 & -p \end{pmatrix} r \right]_{@r=KR} d\theta \, dz$$

$$= \int_0^L \int_0^{2\pi} \begin{pmatrix} -p \\ 0 \\ \mu \frac{\partial v_z}{\partial r} \end{pmatrix}_{r\theta z} \Bigg|_{@r=KR} KR \, d\theta \, dz$$

$$= \int_0^L \int_0^{2\pi} \left(-p \, \hat{e}_r + \mu \frac{\partial v_z}{\partial r} \hat{e}_z \right) \Big|_{@r=KR} KR \, d\theta \, dz$$

$$= \int_0^L \int_0^{2\pi} (-p \, \hat{e}_r) \Big|_{@r=KR} KR \, d\theta \, dz + \int_0^L \int_0^{2\pi} \mu \frac{\partial v_z}{\partial r} \Big|_{@r=KR} \hat{e}_z \, KR \, d\theta \, dz$$

$$v_z = V \left[\frac{\ln(r/R)}{\ln(K)} \right] \rightarrow \text{solved from previous part}$$

$$\frac{\partial v_z}{\partial r} = \frac{V}{\ln K} \left(\frac{R}{r} \right) \cdot \frac{1}{R} = \frac{V}{\ln K} \frac{1}{r}$$

↳ @ $r = KR$ → $\frac{\partial v_z}{\partial r} = \frac{V}{\ln K} \left(\frac{1}{KR} \right) \neq f(r, \theta, z)$

$$\therefore (KRL) \int_0^{2\pi} (-p \, \hat{e}_r) \Big|_{@r=KR} d\theta + L \int_0^{2\pi} \left(\frac{\mu V}{\ln K} \cdot \frac{1}{KR} \cdot KR \right) \hat{e}_z \, d\theta$$

$\hat{e}_r$ → unit vector but has some θ dependence

↳ express in θ units : $\hat{e}_r = \cos\theta \, \hat{e}_x + \sin\theta \, \hat{e}_y$

$$= (KRL) \int_0^{2\pi} -p_{KR} \cdot \begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix}_{xyz} d\theta + \frac{\mu V L}{\ln K} (2\pi) \, \hat{e}_z$$

$\int_0^{2\pi} \cos\theta \, d\theta = \int_0^{2\pi} \sin\theta = 0$; $\hat{e}_z$ independent of θ direction

Centre term goes to zero!)

$$\underline{F} = \frac{\mu V L (2\pi)}{\ln K} \hat{e}_z = \begin{pmatrix} 0 \\ 0 \\ \end{pmatrix}$$

centre term goes to zero :)

$$\underline{F} = \frac{\mu V L (2\pi)}{\ln k} \hat{e}_z = \begin{pmatrix} 0 \\ 0 \\ \frac{\mu V L (2\pi)}{\ln k} \end{pmatrix}_{r\theta z}$$

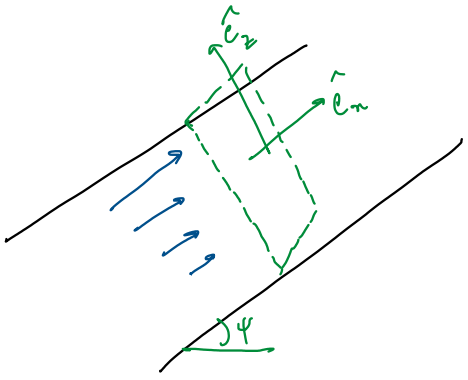
→ Always a vector!

→ physically: + longer the wire, longer the force required to yank it out
+ Faster the V, more force exerted on the wire surface

Makes sense!

6.39

Sunday, 29 October 2023 6:50 PM



$$\frac{\partial P}{\partial z} = -\rho g_z$$

$$P = -\rho g_z z + f(r) \quad ; \quad \frac{dP}{dr} = \frac{\Delta P}{L} = \frac{df}{dr}$$

$$f(r) = \frac{\Delta P}{L} r + C_1$$

$$P = -\rho g_z z + \frac{\Delta P}{L} r + C_1$$

BC → $P(z, r) = P(0, 0) = P_0 \} \Rightarrow C_1 = P_0$

$$\therefore \boxed{P = -\rho g_z z + \frac{\Delta P}{L} + P_0}$$

$$\frac{1}{r} \frac{\partial}{\partial r}(\rho r v_r) = 0$$

Incomp ✓

$$\Rightarrow \rho \frac{\partial}{\partial r}(r v_r) = 0$$

$\frac{\partial}{\partial r}(r v_r) \neq \frac{\partial v_r}{\partial r}$

↳ $\frac{\partial}{\partial r}(r v_r) \neq \frac{\partial v_r}{\partial r}$

Integrate once

$\Rightarrow \quad r v_r = c_1$

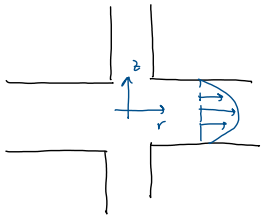
$$\boxed{v_r = \frac{c_1}{r}}$$

$$Q = \langle v \rangle \cdot A$$

7.10

Monday, 6 November 2023 5:47 PM

Radial flow b/w parallel plates



$P(r)$

$v_r(r,z)$

+ Continuity: $v_r = \frac{f(z)}{r}$ (arbitrary function of z (which we don't know!))

+ Navier stokes:

r-component: $\rho\left(v_r \frac{\partial v_r}{\partial r}\right) = -\frac{\partial P}{\partial r} + \mu \frac{\partial^2 v_r}{\partial z^2}$

Although v_r is changing w/ r (as you go further away to the periphery)

We can assume this process is slow enough that only variation in v_r in z-direction dominates

This is the "slow" or developed flow assumption

Now we have:

$$\frac{\partial P}{\partial r} = \mu \frac{\partial^2 v_r}{\partial z^2}$$

We have said $\frac{\partial v_r}{\partial r} \approx 0$ but that does not mean $v_r \neq f(r)$

Cannot solve this expression as of yet since v_r is $f(r,z)$ so cannot take integral of both sides!

However, substitute $v_r = \frac{1}{r} f(z)$

$$\frac{dP}{dr} = \mu \frac{1}{r} \frac{d^2 f}{dz^2}$$

$$r \frac{dP}{dr} = \mu \frac{d^2 f}{dz^2} = \lambda$$

Now separation of variables is possible! (since f is only function of z & P only of r)

λ: Common parameter

$r \frac{dP}{dr} = \lambda$

BC's ✓

↓

solve for λ here

↓

Which we can plug in here at the end

Check OH of Exam 2 review session!

$\frac{d^2 f}{dz^2} = \frac{\lambda}{\mu}$

BC's ✓

↓

Solve for f

↓

get v_r

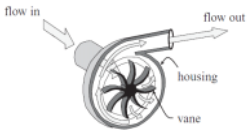
Pump Characteristics Curves

Theory & Background

Pumps → mechanical devices that allow for fluid transportation via pipes (Section 9.2.4.1)

↳ Centrifugal pumps are a common type → transfer rotational energy (from electric motor) to fluid movement

+ Working principle is energy balance! Electric energy → shaft work (by pump to fluid) → affects speed, pressure & other fluid characteristics
+ Pump manufacturers need to characterize a pump → pump operation under various conditions [Pump curves!]



$\Delta E = Q + W$ (W: work done on system by environment)
- ΔE: total energy of system
- Q: Heat from environment to the system via boundary

Energy balance on CV: $\frac{dE_{cv}}{dt} + \int_{cs} (\hat{n}\cdot\underline{v})\rho\hat{E}\,dS = \dot{Q}_{in,cv} - \dot{W}_{by,cv}$ $\quad [E_{cv} = U + E_k + E_p]$

- Closed system
- Open system (mass flow in & out)

↓ Some assumptions

$\Delta E_p + \Delta E_k + \Delta U = Q_{in,cv} - W_{by,cv}$ $\quad (W_f = F\cdot d = (P\cdot A)\cdot d)$

$W_{by,cv}$:
- (W_s) Shaft work (work performed by fluid due to moving parts)
- Flow work (W_f) (work done by fluid as it enters & leaves the system)

$H = U + PV$ ↓

SS Macro Energy balance:

$$\Delta E_p + \Delta E_k + \Delta H = Q_{in,cv} - W_{s,by\,CV}$$

↓ Single in, single out, Incompressible fluid

$$\dot{m}\left(\frac{\langle v\rangle_2^2}{2} - \frac{\langle v\rangle_1^2}{2}\right) + \dot{m}g(z_2 - z_1) + \dot{m}\left(\hat{U}_2 - \hat{U}_1 + \frac{P_2}{\rho} - \frac{P_1}{\rho}\right) = \dot{Q}_{in,cv} - \dot{W}_{s,by\,CV}$$

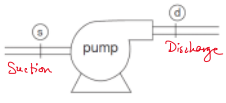
↓ pressure; Temp constant, no phase changes/chemical reaction, negligible heat transfer

Mechanical energy Balance (MEB)

$$\frac{\Delta P}{\rho} + \frac{\Delta\langle v\rangle^2}{2\alpha} + g\Delta z + F = -\frac{W_{s,by}}{m}$$

- ρ: density
- $\langle v\rangle$: average velocity
- $\alpha = 1$ (turbulent flow)
- F: friction term $[\Delta W - Q]$

Pump can add energy via:
+ Increase velocity of fluid
+ Create pressure differential
+ Pump can change fluid height (raise its potential energy)



η = Pump efficiency = $\frac{\text{(useful energy)}}{\text{(total energy)}}$ (energy converted to work (rather than friction))

$= \frac{\left(-\frac{W_{s,by}}{m} - F\right)}{}$

$\left(\frac{-W_{s,bg}}{m}\right)$

Pump brake horsepower: $bhp = -W_{s,bg}$ (Total work delivered by pump)

↓ Dividing by g

$$\therefore \quad \frac{\Delta p}{\rho g} + \frac{\Delta \langle v \rangle^2}{2\alpha g} + \Delta z = \eta \left(\frac{-W_{s,bg}}{mg}\right) = \eta \cdot (bhp)$$

$$\boxed{\left(\frac{P_d}{\rho g} + \frac{\langle v \rangle_d^2}{2g} + z_d\right) - \left(\frac{P_s}{\rho g} + \frac{\langle v \rangle_s^2}{2g} + z_s\right) = \eta \left(\frac{-W_{s,bg}}{mg}\right)}$$

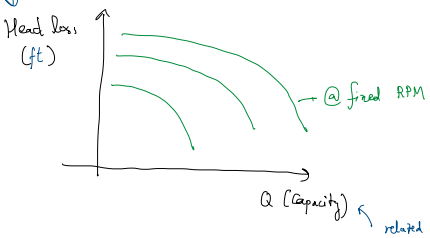
* This form is useful when we know efficiency of pump already (since that takes into account friction loses)

Discharge head (H_d) [=] length

Suction head (H_s) [=] length

$$\underbrace{(H_d - H_s)}_{\text{Head loss}} = \eta \left(\frac{-W_{s,bg}}{mg}\right) = \eta \left[\frac{-bhp}{mg}\right)$$

gives info about η & bhp



Pumping Ability of a CP:
+ Shaft rotational speed (RPM)
+ Shape of vanes
+ flow path

Carrying out the lab:

Characterize relationships b/w discharge flow rate, net head, bhp, efficiency in a CP

↳ Vary Q (via rotameter)
↳ Measure pressure loss → calculate head loss (H_{loss} or $H_{net} = H_d - H_s$)
↳ Non-dimensionalize variables
& then plot them!

Lab Calculations, Discussions & Variables

9.358

$$H_{net} = \text{Head loss} = \frac{P_{out} - P_{in}}{\rho g} + \frac{\langle v \rangle_{out}^2 - \langle v \rangle_{in}^2}{2g\alpha} + \Delta z + \frac{\Sigma}{g}; \qquad F_{piping} = \sum_{j,\text{straight pipe}} \left[4 f_j \frac{L_j}{D_j} \frac{\langle v \rangle_j^2}{2}\right] + \sum_{i,\text{fittings}} \left[n_i k_{f,i} \frac{\langle v \rangle_i^2}{2}\right]$$

where,
f is friction factor given for steady turbulent flow in smooth pipes [9.361]

Non-dimensional variables:

$$C_H = \frac{g H_{net}}{n^2 D^2}$$

n = rotational speed (on pump label)
D = impeller diameter (handout)

$$C_Q = \frac{Q}{nD^3}$$

$$C_P = \frac{bhp}{\rho n^3 D^5}$$

$$\eta = \frac{C_H C_Q}{C_P}$$

+ Plot the characteristic pump curves: (C_H vs. C_Q, C_P vs. C_Q, η vs. C_Q)
(for water & CMC solution)

* can use multiple graphs or one graph w/ secondary axis ↓

Be creative w/ your data graphs to allow readers to understand

+ Discuss the trends! → Why are they like that?
Does it make sense?
How do these compare w/ real pump curve (reference other sources)

+ Error Analysis!
↳ What is error associated w/ all the variables
H_{net}, C_H, C_P, C_Q, η

↳ Are they high/low?
↳ Can they account for the deviations seen in the trends (if any?)?
↳ What are the sources of error in the experiment

Good luck!